Mobile Vaccination Clinics in Northern Nigeria: Addressing Access and Information Barriers

Yousra Hussein, Fabiha Moin, Maren Sautter, Qingfeng Yu

INTRODUCTION

Immunizations are widely regarded as one of the most effective public health interventions, preventing both communicable and non-communicable diseases. However, as of 2024, 14.3 million children worldwide have received no vaccinations at all (World Health Organization, 2025). These so-called zero-dose children are highly vulnerable to preventable diseases and account for nearly half of all vaccine-preventable deaths (Gavi, 2021). Progress toward global immunization targets remains uneven, as several important childhood vaccines fall short of the Immunization Agenda 2030 goals, particularly in sub-Saharan Africa (SSA) (World Health Organization, 2025; Jean Simon et al., 2025).

Sub-Saharan Africa continues to bear a disproportionate burden of under-five mortality and exhibits some of the largest disparities in vaccination coverage worldwide (Bobo et al., 2022; Jean Simon et al., 2025). Across SSA, access to immunization services is often closely linked to socioeconomic status and demographic factors, reinforcing deep-seated health disparities. These patterns suggest that low vaccination uptake reflects deeper structural barriers that limit access to and utilization of preventive care. Addressing the childhood immunization gap and its resulting health and mortality disparities requires a better understanding of the constraints that prevent caregivers from fully immunizing their children, as well as the underlying social determinants of health.

In sub-Saharan Africa, population size and country size negatively affect childhood vaccination coverage (Jean Simon et al., 2025). Nigeria, the largest country in Africa by population, with over 237 million people, has one of the region's lowest full vaccination (FV) coverage for children (Dierks, 2026; Jean Simon et al., 2025). Nationally, only about 31 percent of children aged 12-23 months are fully immunized, while roughly 19 percent are zero-dose children who have received no vaccines (Williams et al., 2024). Coverage disparities are especially pronounced within Nigeria, with routine immunization data showing consistently lower rates in northern states, such as Kano and Kaduna, than in southern states such as Enugu (Williams et al., 2024; Dadari et al., 2023; Ophori et al., 2014). Socioeconomic and demographic factors reinforce these disparities as children from the wealthiest households are substantially more likely to be fully vaccinated than those from the poorest (Williams et al., 2024). Lower vaccination rates are also associated with limited maternal education, maternal age at birth, sex of child, birth order, and reduced engagement with antenatal care services (Eboreime et al., 2015; Williams et al., 2024). These patterns point to the role of both sociocultural and access-related factors in shaping vaccination behavior among caregivers in Nigeria, including regional differences that influence beliefs, norms, and trust in the health system (Jean Simon et al., 2025; Eboreime et al., 2015).

Determinants of Childhood Vaccination Disparities in Nigeria

Existing research highlights two primary perspectives on the disparity in coverage between northern and southern Nigeria. The first and most dominant view is that socio-cultural differences in health-seeking behavior shape vaccination uptake by influencing trust in vaccines and cultural beliefs and practices surrounding childbirth and fertility. The second perspective holds that supply-side factors, particularly access to routine immunization services, drive these disparities.

Ethnicity, religion, and socioeconomic status intersect in complex ways across Nigeria, contributing to regional differences in health-seeking behavior. Northern Nigeria is predominantly Muslim, while the South is largely Christian. Historical and socioeconomic factors, including differences in colonial exposure and access to education, have shaped these regional patterns. In particular, southern

populations tend to have higher levels of educational attainment and economic mobility, which may influence engagement with formal health services, including vaccination (Eboreime et al., 2015). Studying maternal health care utilization among 2,148 mothers in Nigeria, Babalola and Fatusi find substantial regional and ethnic differences, with higher service utilization in the south than in the north (Babalola & Fatusi, 2009). For instance, among the Hausa, a predominantly Muslim ethnic group in northern Nigeria, the cultural concept of *kunya* (shame) plays an important role in childbirth and child-rearing practices and may discourage women from seeking formal health services for themselves or their children (Babalola & Fatusi, 2009). In his study on childhood immunization uptake in Nigeria, Antai similarly found that children born to Igbo mothers, a predominantly Christian ethnic group in southern Nigeria, were more than twice as likely to be fully immunized compared to children of Hausa, Fulani, or Kanuri mothers, who are largely from predominantly Muslim populations in northern Nigeria (Antai, 2009). Antai emphasizes that these differences cannot be attributed solely to cultural or religious factors, as the higher socioeconomic status and economic mobility of Igbo populations may also increase the likelihood of FV among their children (Antai, 2009). Jean Simon and colleagues also document lower FV coverage in Nigeria, which they partly attribute to vaccine hesitancy driven by mistrust in health systems and misinformation, as well as past vaccination campaigns (Jean Simon et al., 2025). Renne similarly argues that some Muslim leaders in northern Nigeria expressed concerns that childhood polio vaccines were contaminated with HIV or anti-fertility agents intended to reduce the Muslim population (Renne, 2006). However, her analysis is based on data from 2005 and may not reflect current attitudes among leaders in the region.

Multiple studies also find that access-related factors, such as distance to vaccination clinics, as well as supply-side constraints including shortages of trained personnel and vaccine availability, significantly affect rates of full childhood immunization. Research by Obanewa and Newell finds that across Nigeria, children whose mothers perceive distance to health facilities as a major barrier have lower full immunization coverage and reduced odds of being fully immunized (Obanewa & Newell, 2020).

Similarly, Eboreime and colleagues find that supply-side issues, such as limited availability of trained vaccinators and uneven geographic coverage, affect childhood vaccination rates in Nigeria, regardless of region (Eboreime et al., 2015). Ophori et al. also identify shortages of vaccines and immunization supplies as a key driver of low childhood immunization rates in Nigeria, and, consistent with prior studies, highlight how Nigeria's siloed federal governance structure can delay the disbursement of health funds to state governments (Ophori et al., 2014).

Overall, the existing literature suggests that childhood vaccination rates in Nigeria are shaped by two interrelated sets of constraints: sociocultural factors, which are deeply related to socioeconomic status, ethnicity, and religious beliefs, as well as access-related factors that determine the availability and rate of utilization of immunization services. These constraints are not mutually exclusive and often reinforce one another, contributing to persistent regional disparities in full vaccination coverage between northern and southern states. As such, improving childhood vaccination rates in Nigeria requires addressing both dimensions. Mobile vaccination clinics, particularly when paired with community-based information and engagement strategies, offer a promising approach by reducing physical access barriers, building trust, and addressing informational gaps at the household level.

BACKGROUND

Economic Epidemiology and Vaccination Uptake

Vaccination uptake can be understood within a broader economic framework. Vaccine-preventable childhood diseases impose costs not only on affected individuals but also on the broader population by increasing transmission risk and necessitating costly self-protection. The SIR (Susceptible, Infected, Recovered) model in health economics provides a useful framework for understanding these dynamics, highlighting how vaccination reduces disease transmission at the population level (Bhattacharya et al., 2022, pp. 454-461). Because the social benefits of vaccination

exceed the private benefits to individuals, the market equilibrium level of vaccination is suboptimal (Bhattacharya et al., 2022). Vaccination generates positive externalities through herd immunity, providing a strong economic justification for public intervention. In theory, Pigouvian subsidies, such as free or incentivized vaccination programs, can help align private incentives with social benefits by encouraging individuals to vaccinate. However, financial subsidies alone are not always sufficient in achieving optimal vaccination coverage. Demand for vaccines is shaped not only by monetary costs but also by non-financial factors, including concerns about side effects, lack of information, and low perceived disease risk (Bhattacharya et al., 2022).

In practice, interventions such as conditional cash transfer programs tied to childhood immunization, alongside free vaccine provision, have been widely used to increase uptake (Connor et al., 2020; Bhattacharya et al., 2022). Yet, as shown in our literature review, Nigeria's low FV coverage is driven by constraints that transcend financial barriers. Informational asymmetries may limit caregivers' understanding of the benefits and timing of immunization, particularly in contexts where mistrust in health systems and misinformation persist. At the same time, supply-side constraints, including distance to health facilities, weak infrastructure, and shortages of trained personnel, can impose significant barriers to accessing services. These demand-side and supply-side constraints suggest the need for interventions that address both dimensions simultaneously. Mobile vaccination clinics can be understood as a form of Pigouvian subsidy that reduces the non-financial costs of accessing care, while, when paired with community-based outreach and information provision, they may also help address informational and trust-related barriers.

Mobile Vaccination Clinics

There is some limited research on implementing mobile vaccination clinics in Nigeria. They are frequently focused on targeting barriers to access by reducing travel time and improving service availability. A 2023 study targeted young, working mothers in markets in Ibadan, southwest Nigeria, providing "friendly immunization, counselling services, and information around vaccination"

(Omobowale et al., 2023). Another recent study targeted mothers of school-aged girls to increase HPV vaccination uptake in Lagos, southwestern Nigeria (Okunade et al., 2024). However, these interventions often do not fully address underlying sociocultural barriers to care, such as mistrust or lack of information, and tend to treat them as secondary considerations. Although studies by Omobowale et al. and Okunade et al. demonstrate the effectiveness of mobile and multisectoral vaccination strategies, the broader literature suggests that optimizing vaccination uptake requires pairing these approaches with proactive outreach and community-driven engagement efforts (Pavoncello et al., 2025; Ophori et al., 2014).

Justifying the Present Study

Routine immunization performance in Nigeria is markedly uneven across regions, with the sharpest gaps concentrated in the North. Northern Nigeria faces both supply and demand constraints, strengthening the case for testing an intervention that bundles improvements in vaccine access with improvements in trust and knowledge systems. A survey study in Zamfara State, northwestern Nigeria, notes that while many attempts have been made to improve immunization, population preferences for interventions have not been documented (Ozawa et al., 2018). The study finds that parents place a high value on door-to-door vaccination, the involvement of religious leaders, media-based information, and strengthening government health services (Ozawa et al., 2018). The same paper also highlights that low trust, particularly stemming from concerns around the polio vaccine, undermines the effectiveness of government, local, and international efforts to expand the supply of childhood vaccines (Ozawa et al., 2018). This underscores the need to experimentally test a mobile health clinic intervention that addresses both the supply and demand sides of childhood immunization.

Some of the strongest causal evidence of mobile vaccination delivery comes from cluster-randomized field experiments showing that increasing access to vaccination points can rapidly increase uptake. A prominent example is a cluster-randomized controlled trial (RCT) across 150 communities in Sierra Leone, which, like Nigeria, is predominantly Muslim, that delivered vaccines and

health professionals to remote communities via mobile vaccination teams, preceded by community engagement efforts. Evidence from the cluster RCT showed that vaccination rates in treatment villages increased by about 26 percentage points in response to this intervention (Meriggi et al., 2024). Additionally, auxiliary populations also utilized these mobile vaccination centers, more than doubling total vaccination counts and suggesting meaningful spillover effects that are highly relevant for northern Nigeria (Meriggi et al., 2024). While COVID-era trials are not identical to routine immunization contexts, the mechanism used in this study is similar, and the Sierra Leone experiment explicitly interprets its effect as overcoming access constraints, recommending bundling with other maternal and child health services. Importantly, cluster designs are common in immunization evaluation. A review of childhood vaccination strategies by Oyo-Ita and colleagues reports that 20 of 41 included studies were cluster-randomized, strengthening our confidence in using cluster randomization (Oyo-Ita et al., 2023). The authors also find that immunization outreach, either alone or combined with non-monetary incentives or health education, likely improves full vaccination uptake among children under five years of age, as does the involvement of community leaders in combination with health provider interventions (Oyo-Ita et al., 2023).

EXPERIMENTAL DESIGN

Intervention

The study evaluates four experimental arms, designed to isolate and jointly test the effects of access and information interventions. The arms include:

Control Arm

Villages in this arm continue to receive standard access to existing fixed-facility immunization services. This arm introduces the counterfactual coverage level under status-quo.

Access Treatment Arm

Mobile vaccination clinics via retrofitted vans with portable refrigerators are deployed to deliver vaccines to participating villages on a schedule aligned with the routine immunization calendar (approximately every four weeks). Nurses staffing the mobile clinics are drawn from the existing public health systems in these villages and administer the same vaccines included in Nigeria's routine immunization schedule.

Information Treatment Arm

Community Health Workers, who constitute the majority of PHC workers (Ajisegiri et al., 2023), are sent to selected villages to deliver targeted health education and reminder interventions to caregivers. This includes structured group education sessions at the community level covering the benefits of vaccination, the recommended schedule, and the vaccines included at each visit. It also includes individual-level reminder messages (i.e., delivered via mobile phone) approximately one week before each scheduled vaccination visit.

Combined Treatment Arm

Villages in this arm receive both the mobile clinic and the information intervention simultaneously. Each intervention is designed to target a specific constraint. Mobile clinics reduce the travel distance and time costs that caregivers face when accessing vaccination services at fixed facilities. Whereas information interventions reduce knowledge gaps and behavioral frictions at the household level. The multi-arm design allows us to estimate not only the average treatment effect of each intervention independently, but also to test for complementarity and to draw inferences about which constraint is more binding in this context. This design extends prior work that typically evaluates a single intervention by testing both independent and joint effects.

Enrollment Criteria and Sampling Strategy

The study will be conducted in rural districts of northern Nigeria[1], where immunization coverage remains lowest and access to health services is limited. Districts will be selected in collaboration with

[1] For instance, basic vaccination coverage among children ranged from 20% (17–23%) in the North West region based on 2018 DHS data (Williams et al., 2024).

local health authorities based on immunization gaps, logistical feasibility, and security considerations[2]. Within each village, the study population consists of children aged 0–24 months residing in rural villages within the study catchment areas[3]. The target outcome population is children aged 12–16 months, who will have had the opportunity to complete all five visits of Nigeria's routine immunization schedule (at birth, 6 weeks, 10 weeks, 14 weeks, and 9 months) by the time of endline data collection. In households with multiple eligible children, one child will be randomly selected as the index child for outcome measurement to avoid intra-household correlation bias.

Assignment Protocol

The unit of randomization is the village (or where villages share a catchment area or are in close proximity, a village cluster). Villages are randomly assigned to one of the four experimental arms. To guard against spillovers between treatment and control units, villages are selected and spaced so as to maximize the minimum distance between study villages assigned to different arms. In cases where two villages are in sufficiently close proximity that travel between them is feasible, they are randomized as a pair and assigned to the same arm.

Randomization will be stratified where possible based on baseline characteristics (e.g., geographic region or baseline coverage) to ensure balance across arms. Treatment effects will be estimated using an intention-to-treat (ITT) framework, comparing outcomes in villages assigned to each treatment arm to outcomes in control villages. The regression specification is:

$$Yiv = \beta 0 + \beta 1 Accessv + \beta 2 Infov + \beta 3 Combinedv + \epsilon iv$$

where Yiv represents vaccination outcomes for child i in village v.

[2] High-intensity conflict can lead to facility closures and prevent "outreach" services which is the primary way rural children are reached (Olu et al., 2024; WHO, 2024).

[3] The study utilizes catchment areas as the primary geographic unit, which may not strictly align with local administrative boundaries. To account for potential spillovers in areas with high geographic density, certain villages will be grouped into randomization clusters based on proximity.

DATA AND POWER CALCULATIONS

Data Inputs

Baseline data are drawn from the 2024 Nigeria Demographic and Health Survey (DHS). The analysis sample is restricted to children aged 0–35 months residing in rural villages across three northern zones (North West, North East, and North Central), yielding 7,434 children in 486 clusters. These data provide: (a) Geographic and sampling identifiers: zone (v024), urban/rural residence (v025), and cluster number (v001), used for stratification by geographic region and baseline coverage level. (b) Baseline immunization coverage: cluster-level full vaccination rates, used for stratification and pre-randomization balance verification across the four experimental arms. (c) Household-level covariates: maternal education (v106), household wealth index (v190), and maternal age (v012), included as regression controls in the ITT specification. In addition, GIS data on village locations will be obtained from local health authorities to enforce spatial separation between villages assigned to different treatment arms and to identify cases where villages in close proximity should be randomized as a pair.

Outputs

The outputs to verify implementation fidelity for each treatment arm are as follows: (a) Access arm: number of mobile clinic visits conducted per village, number of children vaccinated per visit, schedule adherence to the routine immunization calendar (approximately every four weeks), and cold-chain stock-out events. (b) Information arm: number of community-level group education sessions held by Community Health Workers, caregiver attendance at sessions, and individual SMS reminder messages sent and delivered approximately one week before each scheduled visit. (c) Combined arm: all outputs from both arms above. These data will be collected through service delivery logs maintained by the mobile clinic nurses and Community Health Workers, as well as SMS platform delivery records.

Outcome of Interest

The study population consists of children aged 0–24 months residing in rural villages within the study catchment areas. The target outcome population is children aged 12–16 months at endline, who will have had the opportunity to complete all five visits of Nigeria's routine immunization schedule (at birth, 6 weeks, 10 weeks, 14 weeks, and 9 months).

Primary outcome: Full vaccination status, a binary indicator equal to 1 if the child received all 8 doses: BCG (h2), Pentavalent 1–3 (h51, h52, h53), PCV 1–3 (h54, h55, h56), and Measles 1 (h9). A child is coded as vaccinated for a given dose if the corresponding DHS variable takes a value in {1, 2, 3}.

Secondary outcomes: (a) Vaccination count (0–8): number of doses received, capturing partial improvements. (b) Dose-specific uptake: binary indicators for each of the 8 vaccines, allowing identification of where in the schedule dropout occurs.

Table 1: Summary of Data Inputs, Outputs, and Outcomes

Category	Variable	Source	Level	Role
Input	Zone, urban/rural, cluster	DHS 2024	Cluster	Stratification
Input	Baseline full vaccination rate	DHS 2024	Cluster	Stratification, balance
Input	Maternal education, wealth	DHS 2024	Individual	Control variable
Output	Clinic visits, doses given	Service logs	Village	Fidelity check
Output	Sessions held, SMS delivered	CHW/SMS logs	Village	Fidelity check

Outcome	Full vaccination (binary)	Endline survey	Individual	Primary
Outcome	Vaccination count (0–8)	Endline survey	Individual	Secondary
Outcome	Dose-specific uptake	Endline survey	Individual	Secondary

Data Collection Strategy

Data collection involves three sources across different study phases:

Table 2: Data Sources

Data Source	Content	Timing
Nigeria DHS 2024	Baseline vaccination coverage, household characteristics	Pre-intervention (existing)
GIS / local authorities	Village coordinates, inter-village distances for spatial separation	Pre-intervention
Endline household survey	Child vaccination status (card + maternal recall) for children aged 12–16 months	Post-intervention

The endline survey follows DHS methodology. Enumerators visit all villages across four arms and, for each child aged 12–16 months, inspect the vaccination card; if unavailable, maternal recall is recorded. In households with multiple eligible children, one child is randomly selected as the index child for outcome measurement to avoid intra-household correlation bias.

Hypotheses

The ITT regression specification, as stated in the experimental design, is:

$$Y_{iv} = \beta_0 + \beta_1 \cdot Access_v + \beta_2 \cdot Info_v + \beta_3 \cdot Combined_v + X_{iv}\Gamma + \varepsilon_{iv}$$

where Y_{iv} represents vaccination outcomes for child i in village v; $Access_v$, $Info_v$, and $Combined_v$ are mutually exclusive village-level treatment indicators; X_{iv} includes stratum fixed effects and baseline covariates; standard errors are clustered at the village level.

H1 (Access): $\beta_1 > 0$: mobile clinics reduce travel distance and time costs that caregivers face, increasing full vaccination coverage.

H2 (Information): $\beta_2 > 0$: health education sessions and SMS reminders reduce knowledge gaps and behavioral frictions at the household level, increasing coverage.

H3 (Complementarity): $\beta_3 > \beta_1 + \beta_2$: the combined effect exceeds the sum of individual effects, indicating that access and information constraints are complementary.

H4 (Binding constraint): Compare β_1 vs β_2 to assess which constraint is more binding in this context.

Power Calculations (Setup): Baseline Parameters from DHS

The following parameters, estimated from the 2024 Nigeria DHS restricted to rural areas in northern Nigeria (N = 7,434 children in 486 clusters), serve as inputs to the power calculation:

Table 3: Baseline Parameters

Parameter	Value	Source
Baseline full vaccination rate (p_0)	22.4%	DHS 2024 (8 doses, northern rural)
Intra-cluster correlation (ICC)	0.164	One-way ANOVA on fully_vacc by cluster
Average cluster size (m)	15	DHS cluster median
Significance level (α)	0.05	Two-sided test
Statistical power ($1-\beta$)	0.80	Standard threshold

Region-specific baseline rates: North West 18.0%, North East 30.3%, North Central 21.0%.

Sample Size Calculations

Because the unit of randomization is the village (or village cluster where villages share a catchment area), we must account for within-village correlation. The required number of villages per arm is computed in two steps:

Step 1: Individual-level sample size (ignoring clustering). $n_ind = (z_\{\alpha/2\} + z_\beta)^2 \times [p_0(1-p_0) + p_1(1-p_1)] / MDE^2$, where $z_\{\alpha/2\} = 1.96$, $z_\beta = 0.84$, $p_0 = 0.224$, $p_1 = p_0 + MDE$. For an MDE of 10 percentage points ($p_1 = 0.324$), this yields $n_ind \approx 302$ children per arm.

Step 2: Adjust for clustering. $DEFF = 1 + (m - 1) \times ICC = 1 + 14 \times 0.164 = 3.30$. Villages per arm $= n_ind \times DEFF / m = 302 \times 3.30 / 15 \approx 68$.

The design effect of 3.30 reflects the correlation in vaccination outcomes among children within the same village (ICC = 0.164), requiring approximately 3.3 times more observations than simple random sampling to achieve equivalent statistical power.

Sample Size Under Alternative Scenarios

Table 4: Required Sample Size by Expected Effect

Expected Effect	Villages/Arm	Total Villages (4 Arms)	Total Children
8 pp (22.4% → 30.4%)	104	416	6,240
10 pp (22.4% → 32.4%)	68	272	4,080
12 pp (22.4% → 34.4%)	48	192	2,880
15 pp (22.4% → 37.4%)	32	128	1,920

Feasibility

The DHS 2024 sampled 486 rural clusters across northern Nigeria. All scenarios above require fewer than 486 villages, confirming that the proposed four-arm cluster RCT is feasible within this region. We recommend targeting an MDE of 10-12 percentage points (192-272 villages total), balancing statistical power with logistical feasibility given the security considerations and infrastructure constraints noted in the experimental design.

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